

Vulnerability Assessment Report

1st January 2026

System Description

The server hardware consists of a powerful CPU processor and 128GB of memory. It runs on the latest version of Linux operating system and hosts a MySQL database management system. It is configured with a stable network connection using IPv4 addresses and interacts with other servers on the network. Security measures include SSL/TLS encrypted connections.

Scope

The scope of this vulnerability assessment relates to the current access controls of the system. The assessment will cover a period of three months, from June 2026 to August 2026. [NIST SP 800-30 Rev. 1](#) is used to guide the risk analysis of the information system.

Purpose

Consider the following questions to help you write:

- *How is the database server valuable to the business?*
 - **The database server is a critical asset to the business because it stores large quantities of necessary data for development and business operations. The business could not operate remotely or efficiently without such a database.**
- *Why is it important for the business to secure the data on the server?*
 - **It is important to secure this data because it might contain PII, SPII, business operations/plans, and applications or software that could be the target of a cyber attack. This information holds significant value for this business and all of its users, and it is necessary to protect for the sake of all who are associated with the business in some way.**
- *How might the server impact the business if it were disabled?*
 - **In the event of an attack that results in the server being taken offline, normal business operations would be disrupted for employees and customers engaging with the business. This disruption could damage the businesses reputation, compromise sensitive information, or cause the value of the business to decrease if not promptly resolved.**

Risk Assessment

Threat source	Threat event	Likelihood	Severity	Risk
<i>Disgruntled employee</i>	<i>Obtain sensitive information via knowledge of database and security controls.</i>	3	3	9
<i>Hacker/ Hactivist</i>	<i>Obtain sensitive information or disrupt server functionality and integrity.</i>	2	3	6
<i>Physical Storage Device</i>	<i>Physical hardware malfunction leading to server downtime and halted business operations.</i>	1	2	2

Approach

Risks considered the data storage and management methods of the business. The likelihood of a threat occurrence and the impact of these potential events were weighed against the risks to day-to-day operational needs. As a security professional, I examined these three sources of potential risk to the business because they are some of the most common risk sources, especially considering that the organization's database is still open to the public. Malicious actors who know the availability of data within the system pose a threat to the business by obtaining sensitive information, taking the server offline, or creating pathways for future attacks. I also included physical storage device failure in my risk assessment because although it is rather unlikely, it is important to consider when evaluating a centralized database that is accessed from many different physical locations.

Remediation Strategy

Implementation of authentication, authorization, and auditing mechanisms to ensure that only authorized users access the database server. This includes using strong passwords, role-based access controls, and multi-factor authentication to limit user privileges. Encryption of data in motion using TLS instead of SSL. IP allow-listing to corporate offices to prevent random users from the internet from connecting to the database. Security controls and user permissions must be regularly audited for discrepancies and compliance with security standards. An IDS system would also reduce the severity of risks as they occur by allowing the IT/infosec team ample time to respond to and remediate security breaches within the organization's database.